

Program 2:

Develop a multistage docker for container orchestration

Step 1: create a folder and write three files Dockerfile, [index.js](#) and package.json

```
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/prg2$ cat Dockerfile
FROM node:20-alpine AS builder

WORKDIR /app

COPY package.json ./

RUN npm install

COPY . .

RUN npm run build

FROM node:20-alpine

WORKDIR /app

COPY --from=builder /app /app

EXPOSE 3000

CMD ["node", "index.js"]
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/prg2$ cat index.js
const express = require('express');
const app = express();
const PORT = 3000;

app.get('/', (req, res) => {
  res.send('Hello from multi-stage Docker!');
});

app.listen(PORT, () => {
  console.log(`Server running on port ${PORT}`);
});
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/prg2$ cat package.json
{
  "name": "program-2",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "start": "node index.js",
    "build": "echo Building the project..."
  },
  "dependencies": {}
}
```

Step 2: install the node js packages

npm init -y

npm install

Step 3: build the image and run the container

```
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530: ~/Dockerfiles/Program2
=> => transferring dockerfile: 256B
=> [internal] load metadata for docker.io/library/node:20-alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df281f4b7674a2d85071aae2af020ffa39f0a770265435
=> [internal] load build context
=> => transferring context: 41.44kB
=> CACHED [builder 2/6] WORKDIR /app
=> CACHED [builder 3/6] COPY package.json ./
=> CACHED [builder 4/6] RUN npm install
=> CACHED [builder 5/6] COPY . .
=> CACHED [builder 6/6] RUN npm run build
=> CACHED [stage-1 3/3] COPY --from=builder /app /app
=> exporting to image
=> => exporting layers
=> => writing image sha256:2aaed73b82ae732d03f8e658addff8c077bb49ddd41828e47e8a7f75bb35938
=> => naming to docker.io/library/program-2
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/Program2$ docker run -d -p 3000:3000 --name node1 program-2
docker: Error response from daemon: Conflict. The container name "/node1" is already in use by container "0618774ed6f0870d18d0e5be917ca99fdb12c1d02cae0e87638441c915141303". You have t
remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/Program2$ docker run -d -p 3000:3000 --name node2 program-2
b50f46b811e77e4bc6c70611092cfa8d5cd045cc4f57565cf7b698d7366f7908
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint node2 (10782cbcf34c71176fa40d5a477f21ad79ebe77c4
a2fbbd81aaa70d117c0970): Bind for 0.0.0.0:3000 failed: port is already allocated

Run 'docker run --help' for more information
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/Program2$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED      STATUS      PORTS                               NAMES
0618774ed6f0   program-2 "docker-entrypoint.s..." 9 minutes ago Up 9 minutes 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp node1
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/Program2$ docker stop 0618774ed6f0
^C1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/Program2$ docker run -d -p 3000:3000 --name node2 program-2
docker: Error response from daemon: Conflict. The container name "/node2" is already in use by container "b50f46b811e77e4bc6c70611092cfa8d5cd045cc4f57565cf7b698d7366f7908". You have t
remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/Program2$ docker run -d -p 3000:3000 --name node3 program-2
e08c6ccdbbe65045ad6ff82aff0664d1c32afec9d802e6d2a754df6adcc0e6d
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/Program2$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED      STATUS      PORTS                               NAMES
e08c6ccdbbe6   program-2 "docker-entrypoint.s..." 3 seconds ago Up 2 seconds 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp node3
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/Program2$
```